

# Report: Optimising NYC Taxi Operations

Include your visualizations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

## 1. Data Preparation

### 1.1. Loading the dataset

#### 1.1.1. Sample the data and combine the files

##### Objective

The dataset consists of twelve monthly Parquet files containing yellow taxi trip records for the year 2023. Since each file contains millions of records, loading the complete dataset at once would require significant memory and increase processing time. Therefore, a sampling approach was adopted to create a representative dataset while ensuring efficient analysis.

##### Methodology

Loaded all twelve monthly Parquet files individually.

Extracted an equal sample from each month's dataset to ensure balanced representation throughout the year.

Combined the sampled records from all months into a single DataFrame.

Used the consolidated dataset for all subsequent data cleaning and exploratory data analysis.

##### Observation

The sampling approach successfully reduced the overall dataset size while preserving the monthly distribution and seasonal trends. The combined dataset was sufficiently representative of the complete 2023 taxi trip data and allowed efficient execution of the analysis without memory-related issues.

##### Conclusion

Sampling and combining the monthly files created a manageable yet representative dataset for analysis. This approach improved computational efficiency while maintaining the reliability of insights generated during the exploratory data analysis.

## 2. Data Cleaning

### 2.1. Fixing Columns

#### 2.1.1. Fix the index

##### Objective

After combining the sampled datasets, an additional index column was found in the dataset. This column was automatically created during the data loading and merging process and did not contribute any useful information for the analysis.

##### Methodology

Examined the dataset structure to identify unnecessary columns.

Verified that the index column contained only sequential values and had no analytical significance.

Removed the redundant index column from the dataset to improve data quality and readability.

##### Observation

The index column did not contain any meaningful business information and was not required for further analysis. Removing it simplified the dataset and ensured that only relevant variables were retained.

##### Conclusion

Removing the unnecessary index column improved the overall structure of the dataset and made it cleaner for subsequent data cleaning and exploratory data analysis.

#### 2.1.2. Combine the two airport\_fee columns

##### Objective

The dataset contained two separate airport fee columns representing the same type of charge. Maintaining duplicate columns could lead to redundancy and inconsistencies during analysis. Therefore, both columns were consolidated into a single airport fee column.

##### Methodology

Examined both airport fee columns to understand their values and identify any overlap.

Merged the information from both columns into a single Airport\_fee column.

Verified that no valid airport fee information was lost during the consolidation process.

Removed the redundant column after successfully creating the unified airport fee field.

#### Observation

The combined airport fee column retained all relevant information while eliminating duplicate data. This simplified the dataset structure and ensured that airport-related charges were stored consistently in a single variable.

#### Conclusion

Combining the duplicate airport fee columns reduced redundancy, improved data consistency, and made the dataset easier to manage during subsequent cleaning and exploratory data analysis.

## 2.2. Handling Missing ValuesHandling Missing Values

### 2.2.1. Find the proportion of missing values in each column

#### Objective

The objective of this step was to identify the columns containing missing values and calculate the percentage of missing data in each column. Understanding the extent of missing values helps in selecting the most appropriate treatment method before performing exploratory data analysis.

#### Methodology

- Examined all columns in the dataset to identify missing values.
- Calculated the percentage of missing values for each column.
- Reviewed the distribution of missing values across the dataset.
- Identified the variables that required further data cleaning.

#### Observation

The analysis indicated that only a few columns contained missing values, while most variables were complete. The missing value proportions were assessed to determine the most suitable imputation strategy for each affected column.

## Conclusion

Calculating the proportion of missing values provided a clear understanding of the dataset's completeness. This assessment helped in selecting appropriate imputation techniques, thereby improving the quality and reliability of the dataset for subsequent analysis.

### **2.2.2. Handling missing values in passenger\_count**

#### Objective

The objective of this step was to handle missing values in the passenger\_count column while preserving the overall distribution of passenger information. Appropriate imputation was performed to ensure that the missing values did not affect the quality of the analysis.

#### Methodology

- Examined the passenger\_count column to determine the extent of missing values.
- Selected an appropriate imputation technique based on the nature and distribution of the data.
- Replaced the missing values while maintaining the integrity of the dataset.
- Verified that no missing values remained in the passenger\_count column after imputation.

#### Observation

The missing values in the passenger\_count column were successfully handled without significantly altering the overall data distribution. The column became complete and suitable for further exploratory data analysis.

#### Conclusion

Handling missing values in the passenger\_count column improved data completeness and ensured that passenger-related analyses could be performed more accurately and reliably.

### **2.2.3. Handle missing values in RatecodeID**

#### Objective

The objective of this step was to handle missing values in the RatecodeID column by replacing them with the most frequent value (mode), ensuring data completeness while preserving the original distribution of the categorical variable.

#### Methodology

Identified missing values in the RatecodeID column.

Calculated the mode of the RatecodeID column.

Replaced missing values with the mode.

Verified that no missing values remained after imputation.

Observation

The missing values in the RatecodeID column were successfully imputed using the mode. The column became complete without affecting the categorical distribution.

Conclusion

Mode imputation effectively handled the missing values in the RatecodeID column, improving data completeness while maintaining the integrity of the categorical data.

#### **2.2.4. Impute NaN in congestion\_surcharge**

Objective

To handle missing values in the congestion\_surcharge column by replacing them with the median value, ensuring data completeness while preserving the overall distribution.

Methodology

Identified all missing values in the congestion\_surcharge column.

Replaced the missing values with the median congestion surcharge.

Verified that no missing values remained after imputation.

Observation

The missing values in the congestion\_surcharge column were successfully handled using median imputation. The column became complete without significantly affecting the overall data distribution.

Conclusion

Handling missing values in the congestion\_surcharge column improved the completeness and reliability of the dataset, making it suitable for further exploratory data analysis.

### **2.3. Handling Outliers and Standardising Values**

#### **2.3.1. Check outliers in payment type, trip distance and tip amount columns**

## Objective

To identify potential outliers in the payment\_type, trip\_distance, and tip\_amount columns before performing further analysis.

## Methodology

Examined the selected columns for unusual or extreme values.

Used summary statistics and visualizations (such as boxplots) to detect outliers.

Compared the values with the overall data distribution to identify abnormal observations.

## Result

Outliers were observed in the trip\_distance and tip\_amount columns, indicating the presence of unusually high values.

The payment\_type column is a categorical variable and did not contain numerical outliers.

The identified outliers were retained for further evaluation, as they may represent genuine taxi trips.

## Observation

The analysis showed that extreme values mainly occurred in trip\_distance and tip\_amount.

These values could influence statistical measures and model performance if not handled appropriately.

## Conclusion

Outlier detection helped identify unusual observations in the dataset. This step provides a basis for deciding whether to retain, cap, or remove outliers during the data preprocessing stage while preserving the quality of the analysis.

# 3. Exploratory Data Analysis

## 3.1. General EDA: Finding Patterns and Trends

### 3.1.1. Classify variables into categorical and numerical

## Objective

To classify the variables in the dataset into categorical and numerical types for appropriate exploratory data analysis.

## Methodology

Examined the data types of all variables.

Classified categorical variables such as VendorID, payment\_type, and store\_and\_fwd\_flag.

Classified numerical variables such as trip\_distance, fare\_amount, tip\_amount, and total\_amount.

## Result

The variables were successfully classified into categorical and numerical groups. This classification made it easier to apply suitable statistical analysis and visualization techniques.

## Observation

The dataset contained both categorical and numerical variables, each requiring different analytical methods for meaningful insights.

## Conclusion

Classifying variables improved the understanding of the dataset structure and ensured appropriate techniques were used during exploratory data analysis.

### **3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months**

## Objective

To analyze the distribution of taxi pickups across different hours, days of the week, and months.

## Methodology

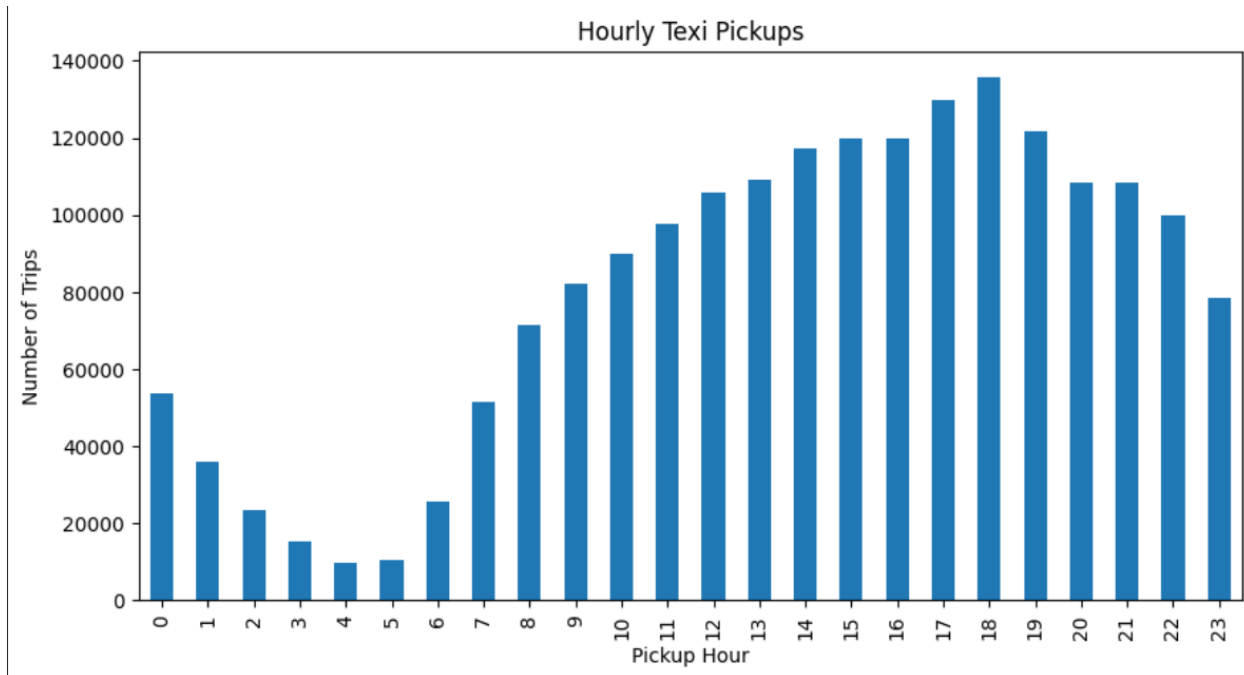
Extracted the pickup hour, weekday, and month from the pickup datetime column.

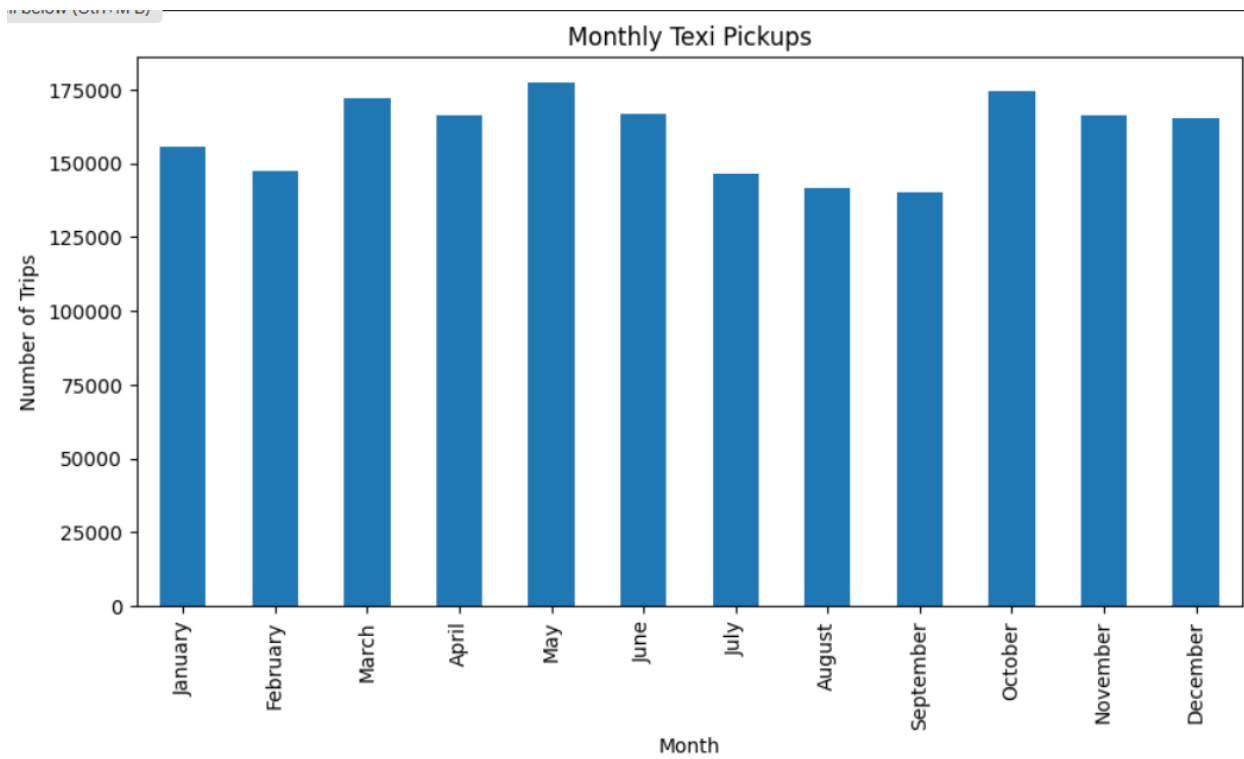
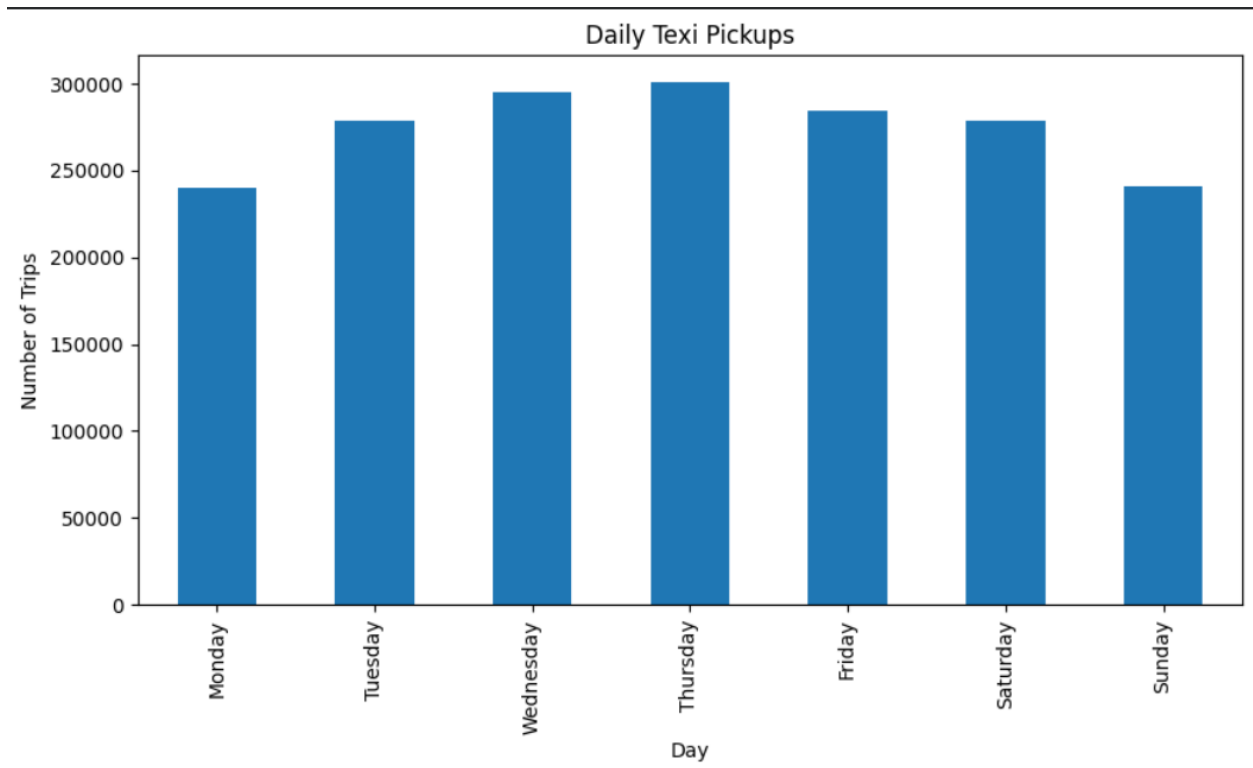
Counted the number of taxi pickups for each time period.

Visualized the distributions using charts.

## Result

The analysis revealed clear variations in taxi pickup demand across different hours, weekdays, and months. Peak pickup activity occurred during the evening hours (6 PM–7 PM), while the lowest demand was observed between 4 AM and 5 AM.





Observation

Taxi pickup demand was not uniformly distributed. Certain hours, weekdays, and months recorded significantly higher trip volumes than others.

#### Conclusion

The analysis identified important temporal patterns that can support better operational planning and demand forecasting.

### **3.1.3. Filter out the zero/negative values in fares, distance and tips**

#### Objective

To identify and remove invalid zero and negative values from the fare, trip distance, and tip amount columns to improve data quality.

#### Methodology

Checked the fare\_amount, trip\_distance, and tip\_amount columns for zero and negative values.

Filtered out invalid records from the dataset.

Verified that the cleaned dataset contained only valid observations.

#### Result

Invalid zero and negative values were successfully removed from the selected columns. The filtered dataset is cleaner and more suitable for further exploratory data analysis.

#### Observation

The dataset contained a small number of invalid records. Removing them improved the accuracy and reliability of the analysis.

#### Conclusion

Filtering zero and negative values ensured better data quality and made the dataset suitable for meaningful analysis and visualization.

### **3.1.4. Analyse the monthly revenue trends**

#### Objective

To analyse the monthly revenue (total\_amount) trend and identify changes in taxi revenue across different months.

## Methodology

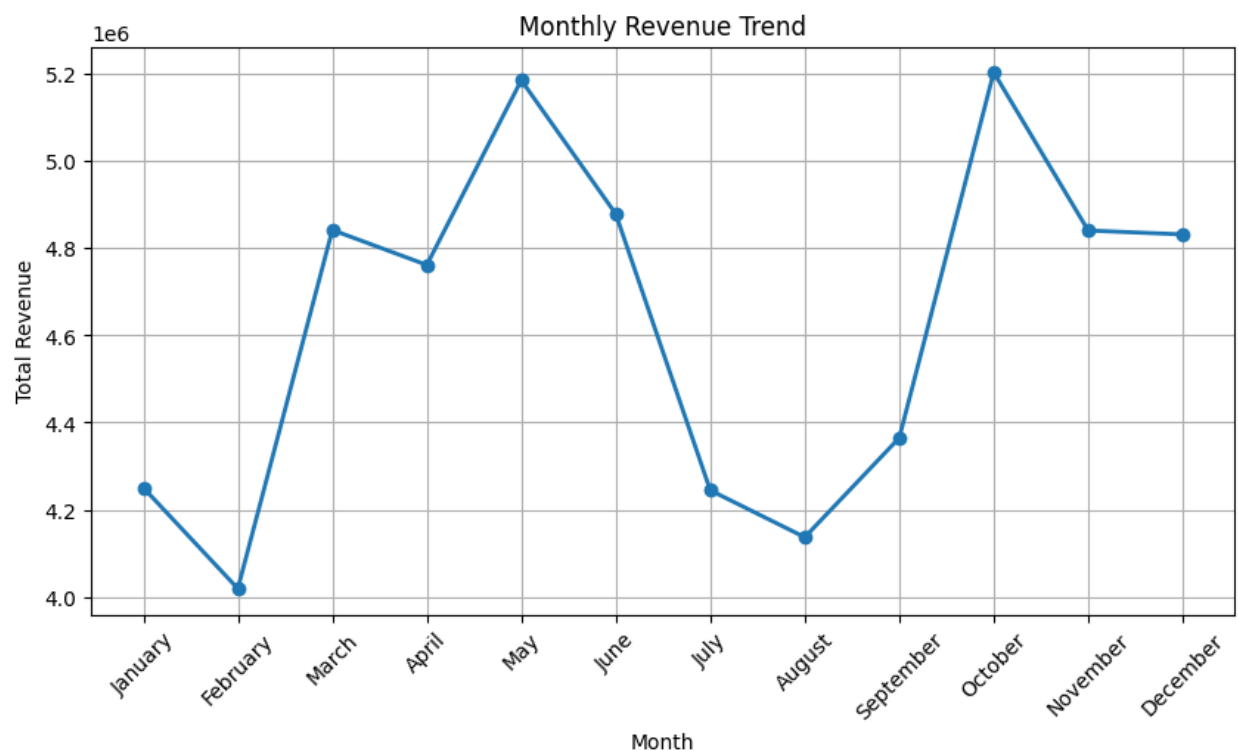
Grouped the dataset by pickup\_month.

Calculated the total monthly revenue using the total\_amount column.

Visualized the monthly revenue trend using a line chart.

## Result

The monthly revenue varied across different months. Some months generated higher revenue, while others showed comparatively lower revenue, indicating seasonal variations in taxi demand.



## Observation

Revenue was not uniform throughout the year. Peak revenue months indicate higher taxi usage, while lower revenue months reflect comparatively reduced demand.

## Conclusion

Monthly revenue trend analysis helps identify seasonal business patterns and supports better operational planning and revenue forecasting.

### 3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

## **Objective**

To calculate the contribution of each quarter (Q1–Q4) to the total yearly revenue and visualize it using a pie chart.

## **Methodology**

Grouped the monthly revenue into four quarters (Q1, Q2, Q3, and Q4).

Calculated the total revenue for each quarter.

Plotted a pie chart to show the percentage contribution of each quarter to the annual revenue.

## **Result**

The quarter-wise revenue distribution was:

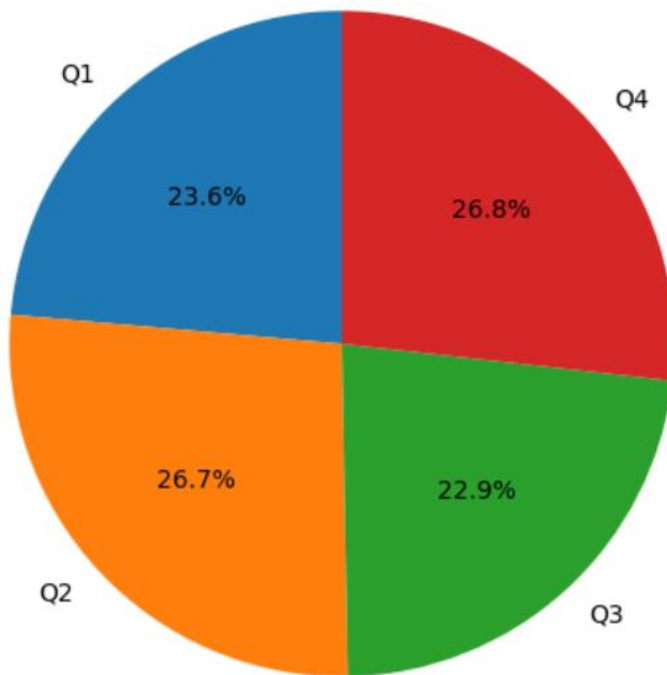
Q1: 23.6%

Q2: 26.7%

Q3: 22.9%

Q4: 26.8%

Quarter-wise Revenue Proportion



#### Observation

Revenue was distributed fairly evenly across the year. However, Q4 contributed the highest share (26.8%), followed closely by Q2 (26.7%), while Q3 had the lowest contribution (22.9%).

#### Conclusion

The yearly revenue was balanced across all four quarters, with a slight increase in revenue during Q2 and Q4. This suggests relatively stable business performance with minor seasonal variations.

### 3.1.6. Analyse and visualise the relationship between distance and fare amount

#### Objective

To analyse the relationship between trip distance and fare amount and calculate the correlation between them.

#### Methodology

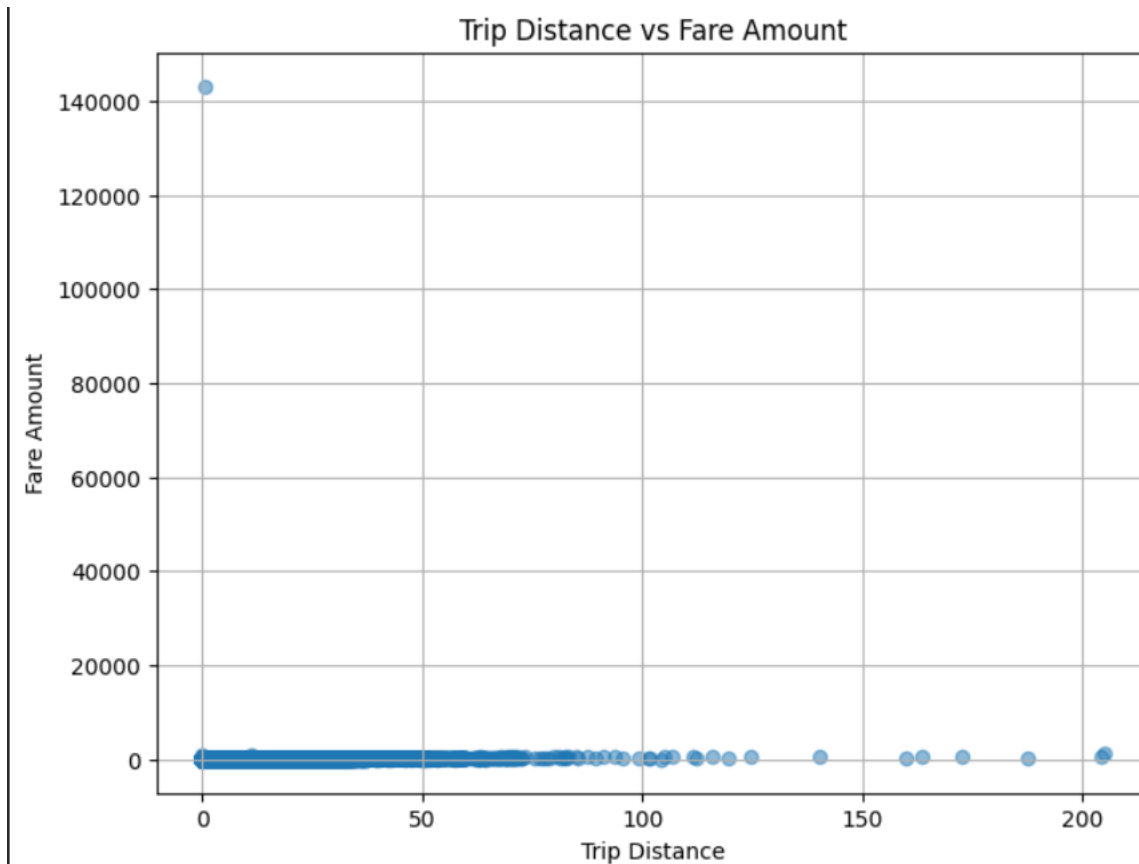
Removed trips with zero trip distance.

Calculated the correlation between trip\_distance and fare\_amount.

Plotted a scatter plot to visualise the relationship.

### Result

The correlation between trip distance and fare amount is 0.16, indicating a weak positive relationship.



### Observation

The scatter plot shows that fares generally increase as trip distance increases, but the relationship is weak due to the presence of outliers and variability in fare values.

### Conclusion

Trip distance has a positive but weak influence on fare amount. Other factors such as tolls, waiting time, surcharges, and traffic conditions also affect the total fare.

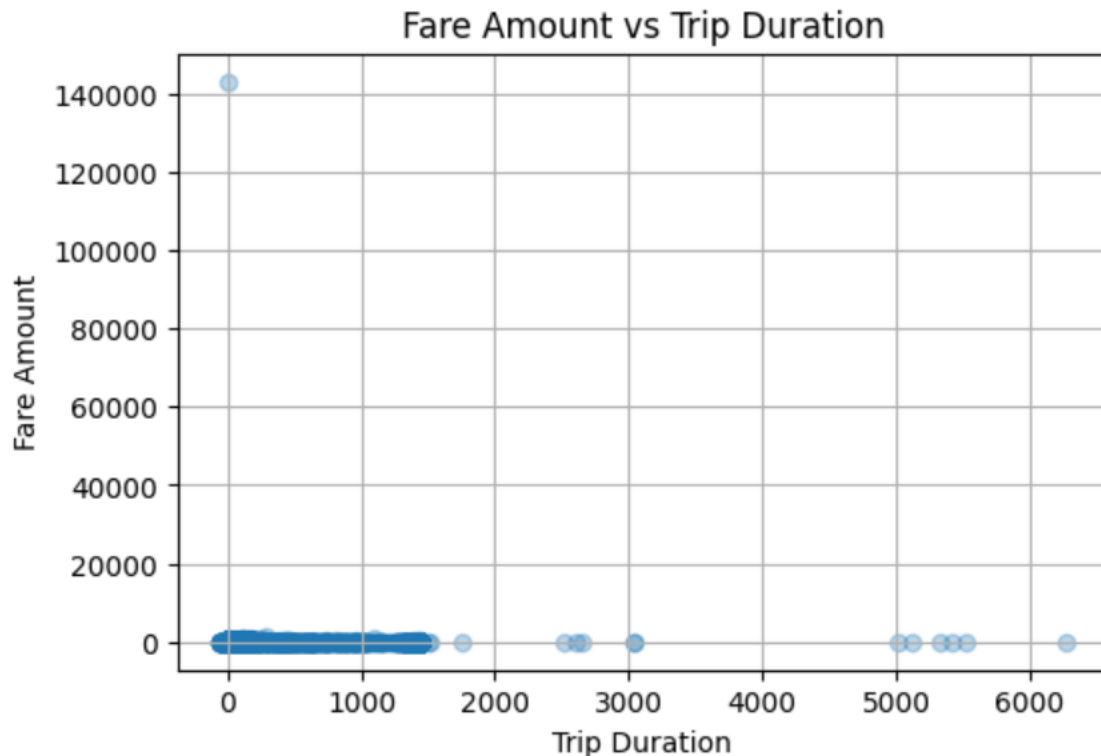
### 3.1.7. Analyse the relationship between fare/tips and trips/passengers

#### 1. Relationship between Fare Amount and Trip Duration

Conclusion: The correlation coefficient (0.045) indicates an extremely weak positive relationship. Therefore, trip duration alone has very little influence on fare amount.

Report statement:

The correlation between fare amount and trip duration is 0.045, which indicates an extremely weak positive relationship. The scatter plot shows no clear linear trend, suggesting that trip duration has little effect on the fare amount. A few outliers with unusually high fares are also observed.

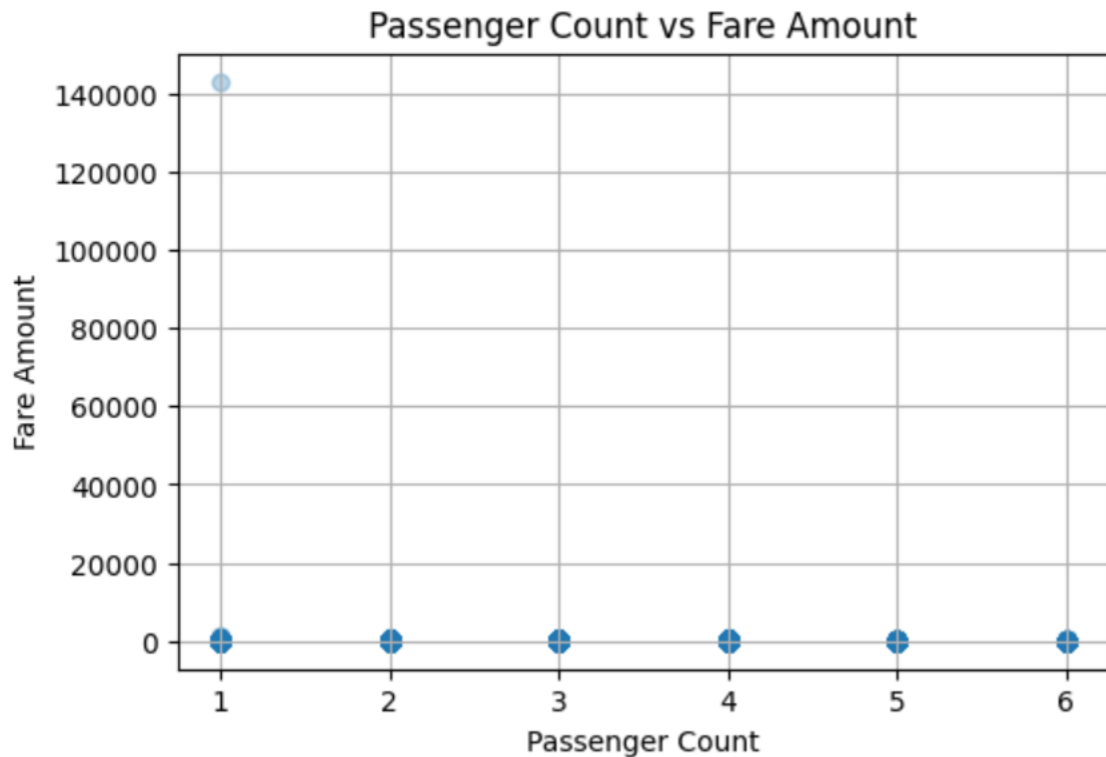


## 2. Relationship between Fare Amount and Passenger Count

Conclusion: The correlation coefficient (0.007) indicates almost no relationship between passenger count and fare amount. Fare values remain similar across different passenger counts.

Report statement:

correlation between fare amount and passenger count is 0.007, indicating virtually no relationship. The scatter plot confirms that fare amounts are similar across different passenger counts, suggesting that the number of passengers has minimal impact on taxi fares.

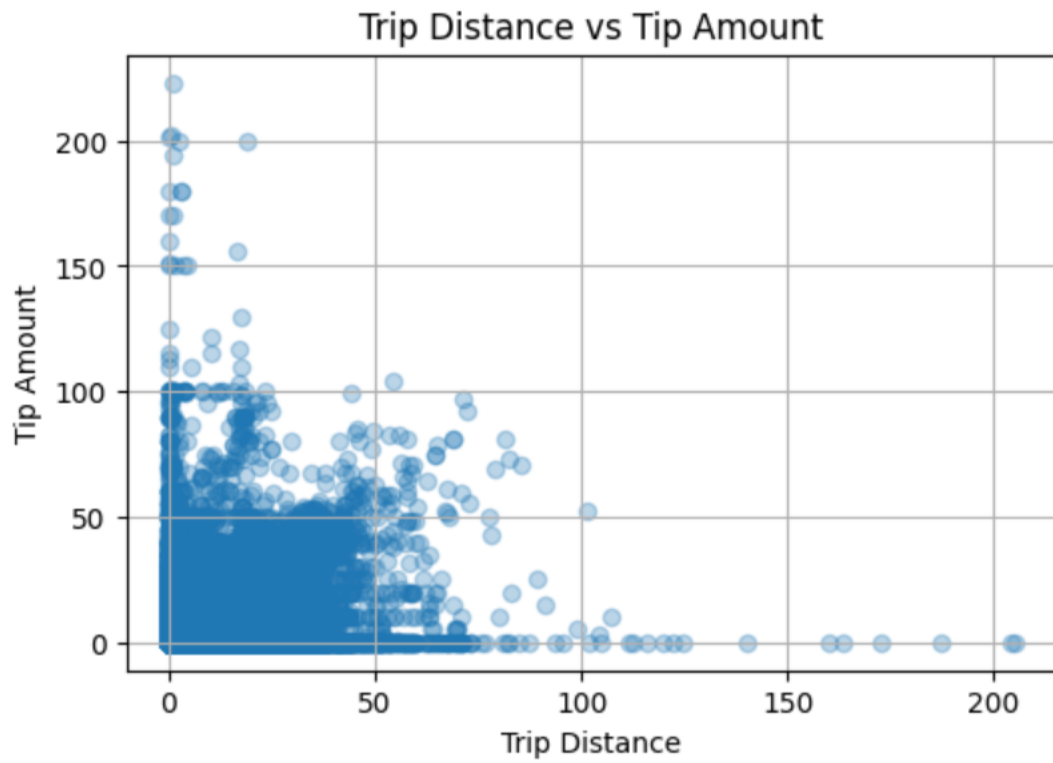


### 3. Relationship between Tip Amount and Trip Distance

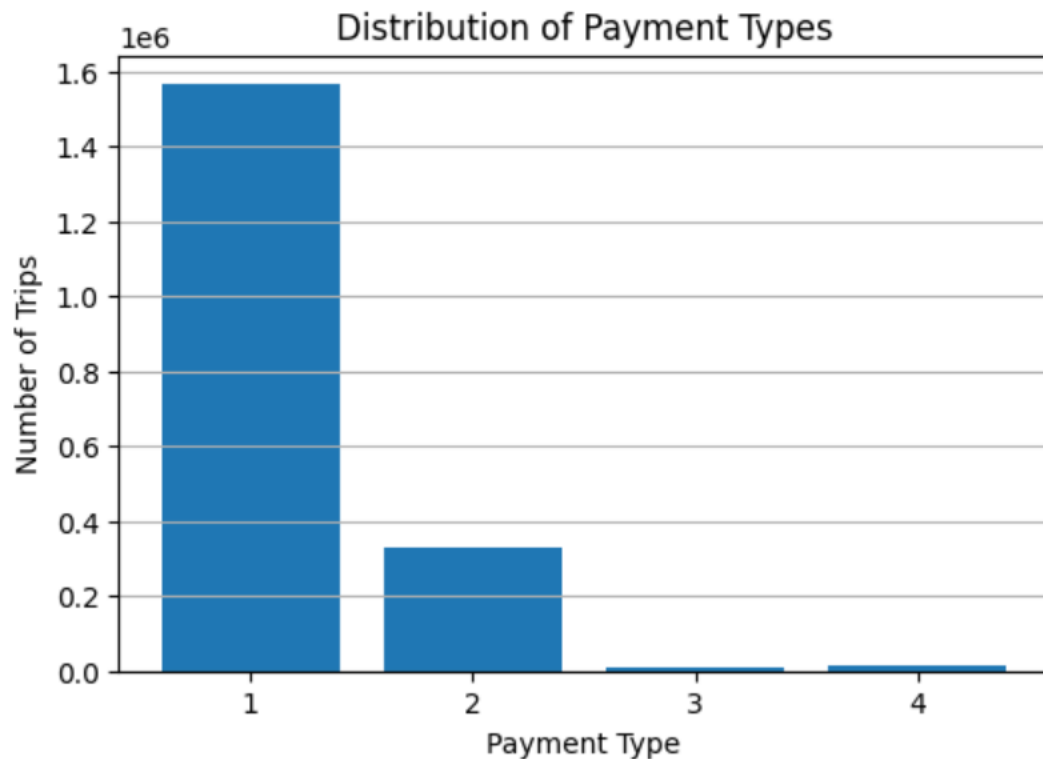
Conclusion: The correlation coefficient (0.572) indicates a moderate positive relationship between trip distance and tip amount. As trip distance increases, passengers generally give higher tips, although some outliers are present.

Report statement:

The correlation between tip amount and trip distance is 0.572, indicating a moderate positive relationship. The scatter plot shows that passengers generally give higher tips on longer trips, although some variation and outliers are present.



**3.1.8. Analyse the distribution of different payment types**



Analysis:

The bar chart shows that Credit Card (Payment Type 1) is the most frequently used payment method. Cash (Payment Type 2) is the second most common payment method. No Charge (Payment Type 3) and Dispute (Payment Type 4) are used very rarely. This indicates that most NYC taxi passengers prefer paying by credit card, while only a small number of trips involve other payment types.

### **3.1.9. Load the taxi zones shapefile and display it**

The taxi zones shapefile was successfully loaded using GeoPandas. The dataset contains information such as OBJECTID, LocationID, zone name, borough, and geometry for each taxi zone. This confirms that the shapefile has been imported correctly and is ready for further geographical analysis.

### **3.1.10. Merge the zone data with trips data**

The trip data was successfully merged with the taxi zone dataset using the PULocationID from the trip records and the LocationID from the zones dataset. This merge added zone-related

information, such as the zone name, borough, and geometry, to the trip records, making the data ready for further location-based analysis.

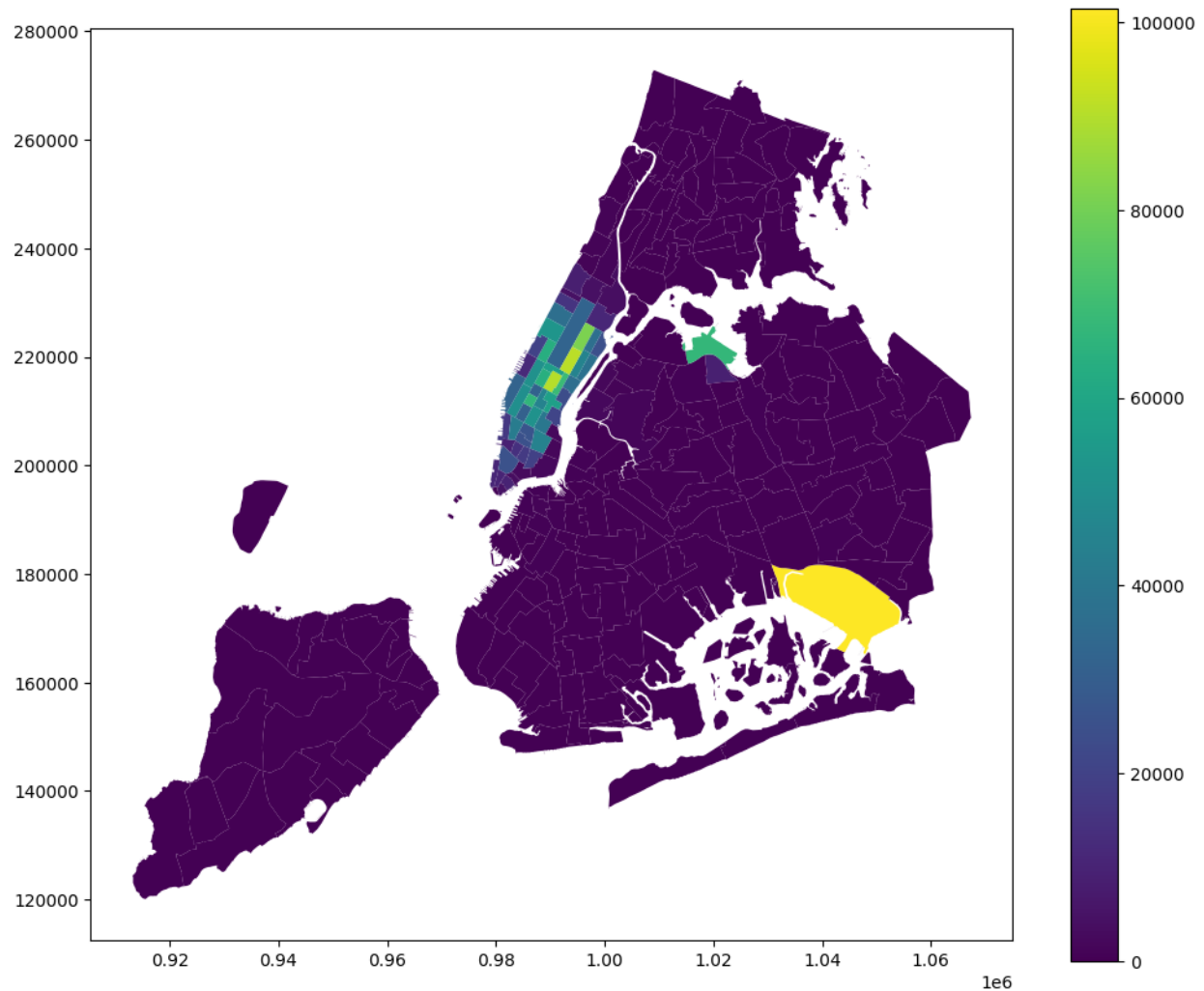
#### **3.1.11. Find the number of trips for each zone/location ID**

The trip data was grouped by PULocationID to calculate the total number of trips for each taxi zone. A new column named `trip_count` was created, which stores the total trips associated with every location ID. This summary helps identify which taxi zones have the highest and lowest trip activity.

#### **3.1.12. Add the number of trips for each zone to the zones dataframe**

The trip counts calculated for each pickup location were merged back into the taxi zones GeoDataFrame using the `LocationID` and `PULocationID` columns. Any zones without recorded trips were assigned a value of 0 using the `fillna()` function. As a result, the GeoDataFrame now contains a new `trip_count` column alongside the geographic information, making it suitable for creating a choropleth map that visualizes the distribution of taxi trips across New York City taxi zones.

#### **3.1.13. Plot a map of the zones showing number of trips**



A choropleth (color-coded) map was created using the `trip_count` column in the `GeoDataFrame`. Different colors represent the total number of taxi trips for each zone. Zones with higher trip counts appear in brighter colors, while zones with fewer trips appear in darker colors. This map clearly shows the geographical distribution of taxi trips across New York City and helps identify the busiest taxi pickup zones.

#### 3.1.14. Conclude with results

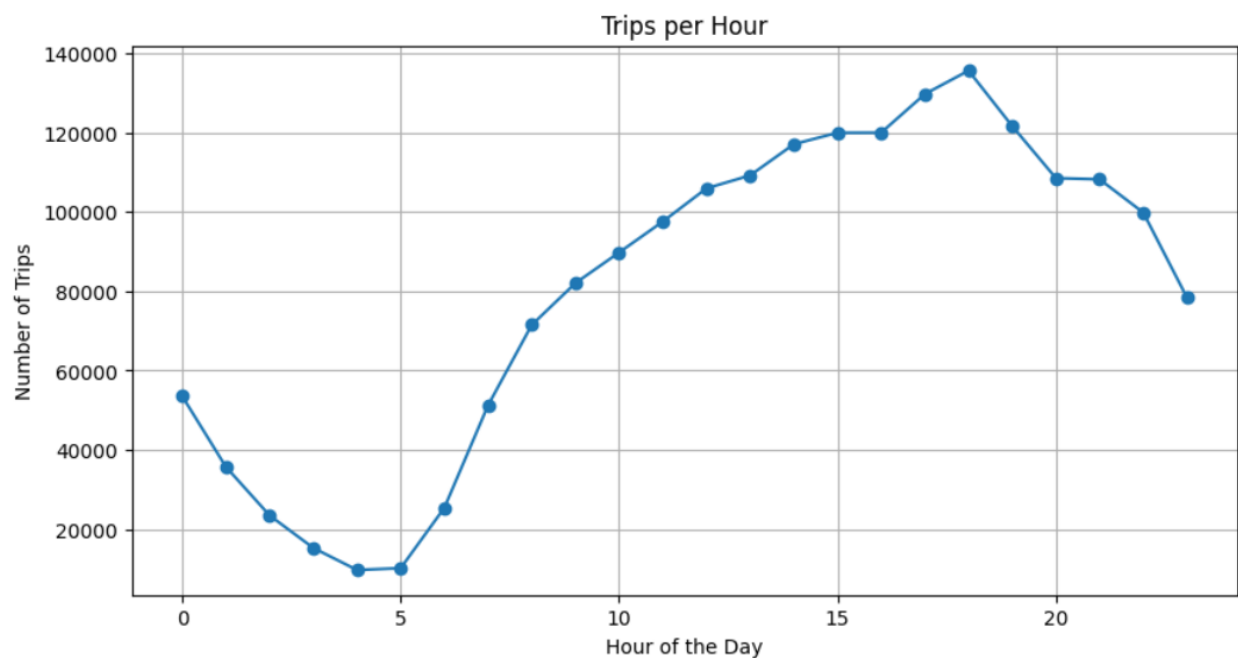
The analysis shows that taxi trips are concentrated in a few major locations. JFK Airport recorded the highest number of trips (101,526), followed by Upper East Side South, Midtown Center, Upper East Side North, and Midtown East. Airport zones and central Manhattan have the highest taxi demand, while many outer borough zones have significantly fewer trips. The choropleth map effectively highlights these differences in taxi activity across New York City.

## 3.2. Detailed EDA: Insights and Strategies

### 3.2.1. Identify slow routes by comparing average speeds on different routes

Observation: The average speed of each route was calculated using the average trip distance and average trip duration. The analysis identified the slowest routes across different hours of the day. These routes have very low average speeds, indicating heavy traffic congestion or frequent stops. Such insights can help taxi operators optimize route planning, reduce travel time, and improve overall operational efficiency.

### 3.2.2. Calculate the hourly number of trips and identify the busy hours



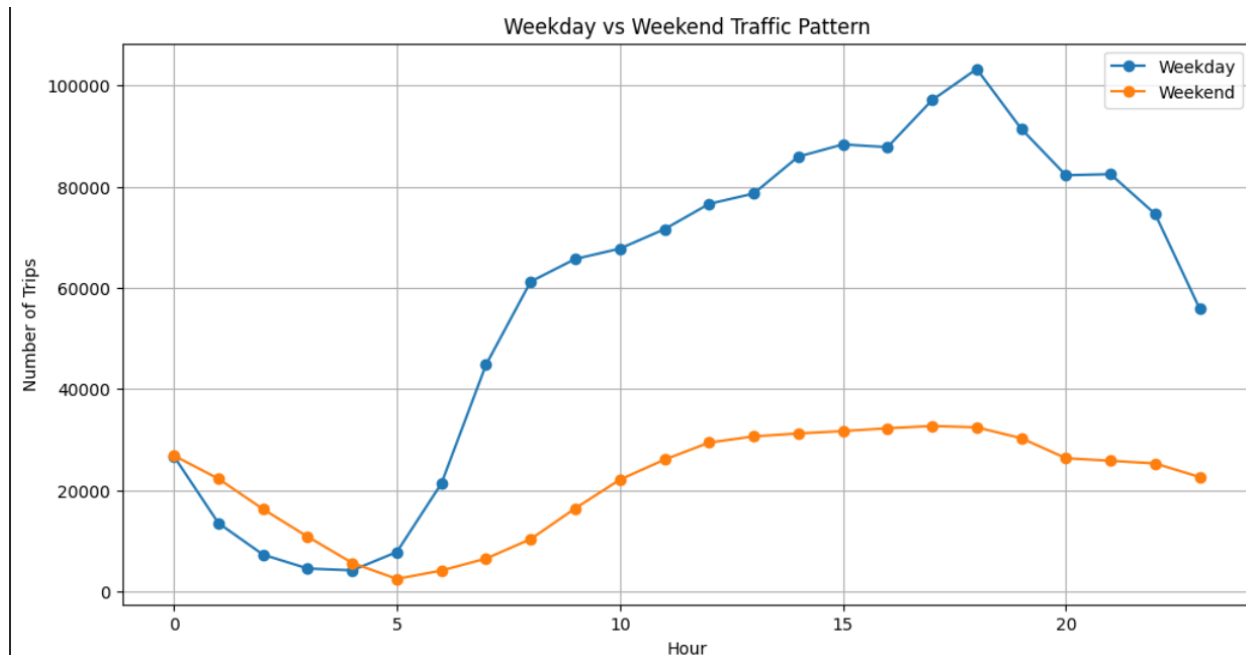
Observation: The hourly trip analysis shows a clear daily travel pattern. Trip demand is lowest during the early morning hours (around 4:00–5:00 AM). Demand increases rapidly after 6:00 AM, reflecting the morning commute, and continues to rise throughout the day. The highest number of trips occurs at 6:00 PM (18:00) with approximately 135,623 sampled trips, after which demand gradually declines during the late evening. This pattern indicates that evening rush hours experience the highest taxi demand.

### 3.2.3. Scale up the number of trips from above to find the actual number of trips

Observation: The sampled trip counts were scaled using the sampling fraction (0.01) to estimate the actual number of taxi trips. The estimated peak demand occurs at 6:00 PM, with approximately 13,562,300 trips, followed by 5:00 PM (12,972,900), 7:00 PM (12,163,800), 4:00

PM (12,000,000), and 3:00 PM (11,997,200). These results confirm that taxi demand is concentrated during the evening commuting period.

#### 3.2.4. Compare hourly traffic on weekdays and weekends



Observation: Weekday and weekend travel patterns differ considerably. Weekday trips increase sharply after the morning hours, reach their highest level during the evening rush period, and remain significantly higher than weekend trips throughout the day. Weekend demand grows more gradually, peaks in the afternoon and early evening, and shows a flatter trend. This indicates that weekday travel is primarily driven by commuting and business activities, whereas weekend travel is associated with leisure and recreational purposes.

#### 3.2.5. Identify the top 10 zones with high hourly pickups and drops

Observation: The analysis identified the busiest pickup and dropoff locations based on trip volume. Pickup Zone 132 recorded the highest number of pickups (101,526 trips), followed by Zones 237, 161, and 236. Similarly, Dropoff Zone 236 recorded the highest number of dropoffs (85,285 trips), followed by Zones 237, 161, and 230. These zones represent major transportation hubs with consistently high passenger activity and should be prioritized for efficient taxi allocation.

#### 3.2.6. Find the ratio of pickups and dropoffs in each zone

Observation: The pickup-to-dropoff ratio was calculated by dividing the total pickups by total dropoffs for each taxi zone. Zone 70 recorded the highest ratio (8.23), followed by Zone 132

(4.62) and Zone 138 (2.89), indicating these areas are primarily passenger origin locations. In contrast, Zones 29, 240, and 171 have ratios close to zero, suggesting they serve mainly as passenger destinations. These findings can assist taxi operators in balancing vehicle distribution and reducing empty return trips.

### **3.2.7. Identify the top zones with high traffic during night hours**

Observation: During nighttime hours (11:00 PM–5:00 AM), Zone 79 recorded the highest number of both pickups (16,266) and dropoffs (8,715), indicating that it remains the busiest location even during late-night periods. Other high-demand pickup zones include 132, 249, 48, and 148, while major nighttime dropoff zones include 48, 170, 68, 107, and 141. These results suggest that nighttime taxi demand is concentrated in a limited number of commercial, entertainment, and residential areas.

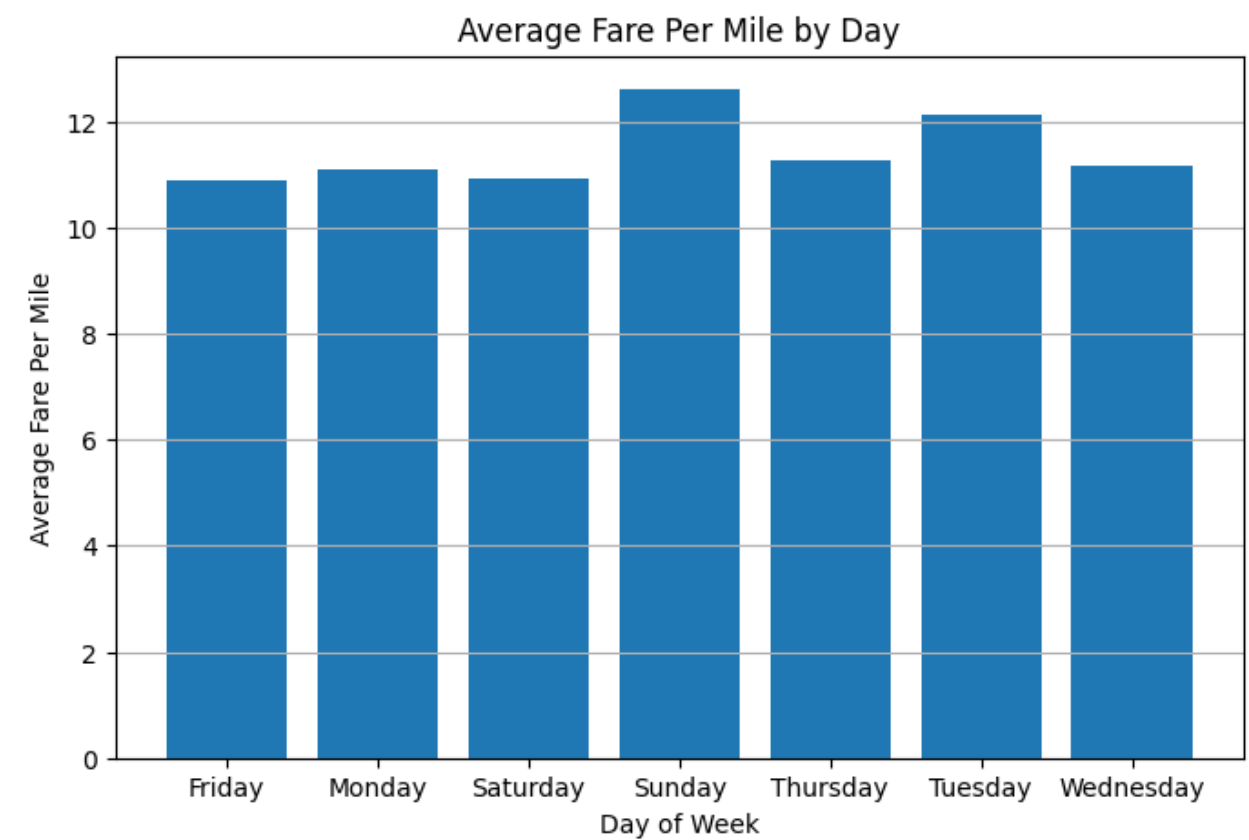
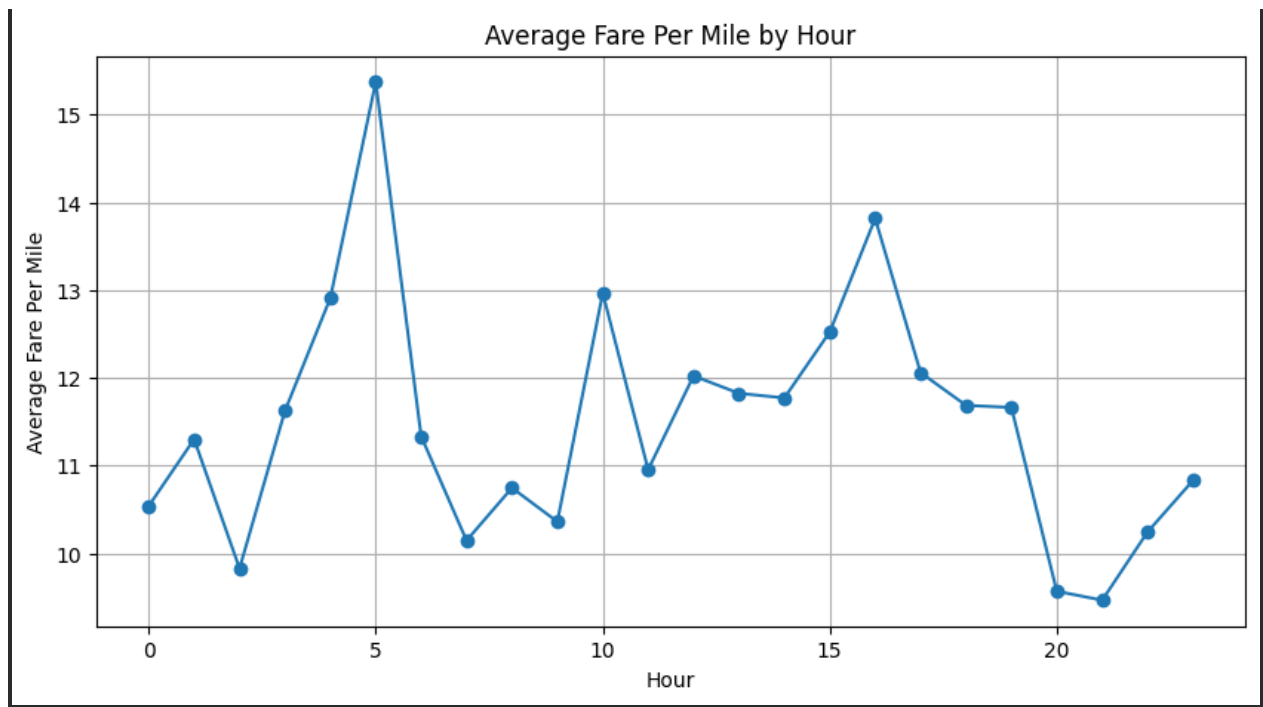
### **3.2.8. Find the revenue share for nighttime and daytime hours**

Observation: The revenue analysis shows that daytime operations (6:00 AM–10:00 PM) generated approximately 88.01% of the total revenue, whereas nighttime operations (11:00 PM–5:00 AM) contributed 11.99%. This indicates that most taxi revenue is earned during daytime hours due to higher passenger demand associated with commuting, business activities, shopping, and tourism.

### **3.2.9. For the different passenger counts, find the average fare per mile per passenger**

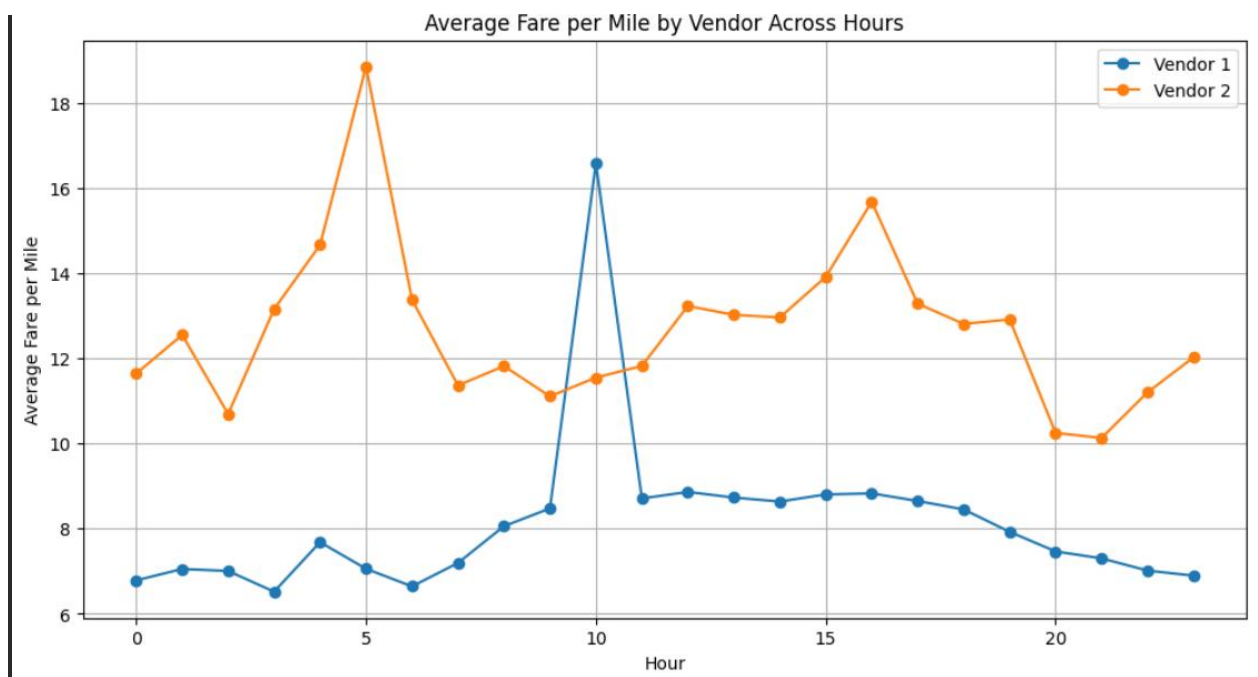
Observation: The analysis shows that the average fare per mile per passenger decreases as the number of passengers increases. Single-passenger trips have the highest fare per passenger, whereas trips with multiple passengers have lower per-person travel costs because the fare is shared among more passengers. This demonstrates the cost efficiency of group travel.

### **3.2.10. Find the average fare per mile by hours of the day and by days of the week**



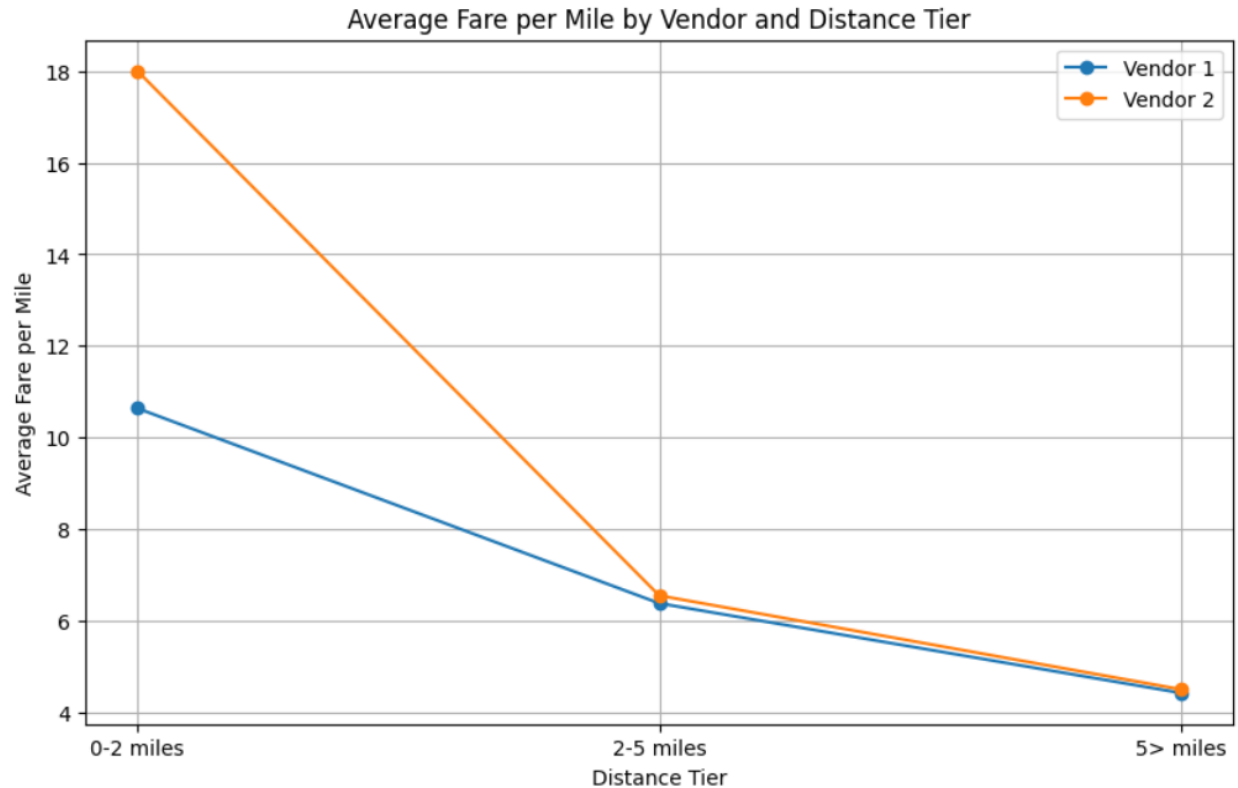
Observation: The average fare per mile varies across different hours of the day. Higher fare rates are observed during certain early morning and evening periods, likely due to increased traffic congestion, longer travel times, or peak-demand pricing. Lower average fares are recorded during off-peak hours. This variation highlights the influence of traffic conditions and travel demand on taxi fare efficiency.

### 3.2.11. Analyse the average fare per mile for the different vendors



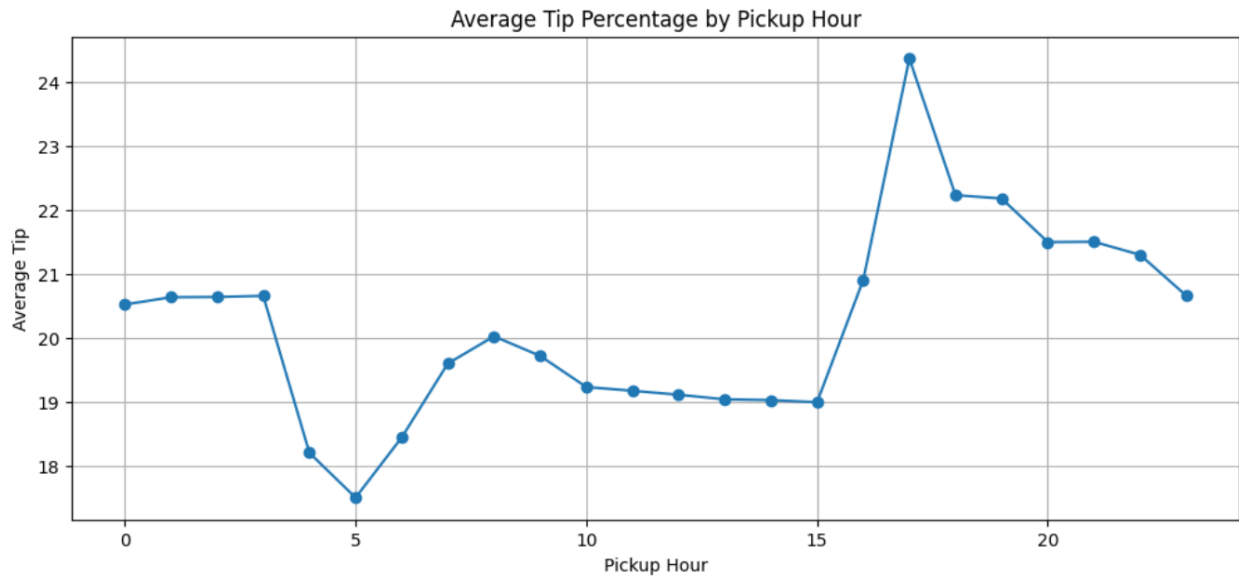
The analysis indicates noticeable differences in the average fare per mile between the two taxi vendors across different hours of the day. Vendor 2 consistently records a higher average fare per mile than Vendor 1 throughout most hours. The highest average fare for Vendor 2 occurs around 5:00 AM, reaching approximately 18–19 USD per mile, after which it gradually decreases. In contrast, Vendor 1 maintains relatively stable fares between 6 and 8 USD per mile, with a temporary spike around 10:00 AM, likely caused by a smaller number of trips or outlier observations. During the afternoon, Vendor 2 continues to have higher fare rates than Vendor 1, while both vendors experience a slight decline during the late evening. Overall, Vendor 2 appears to charge higher fares per mile than Vendor 1 across nearly all hours of the day.

### 3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion



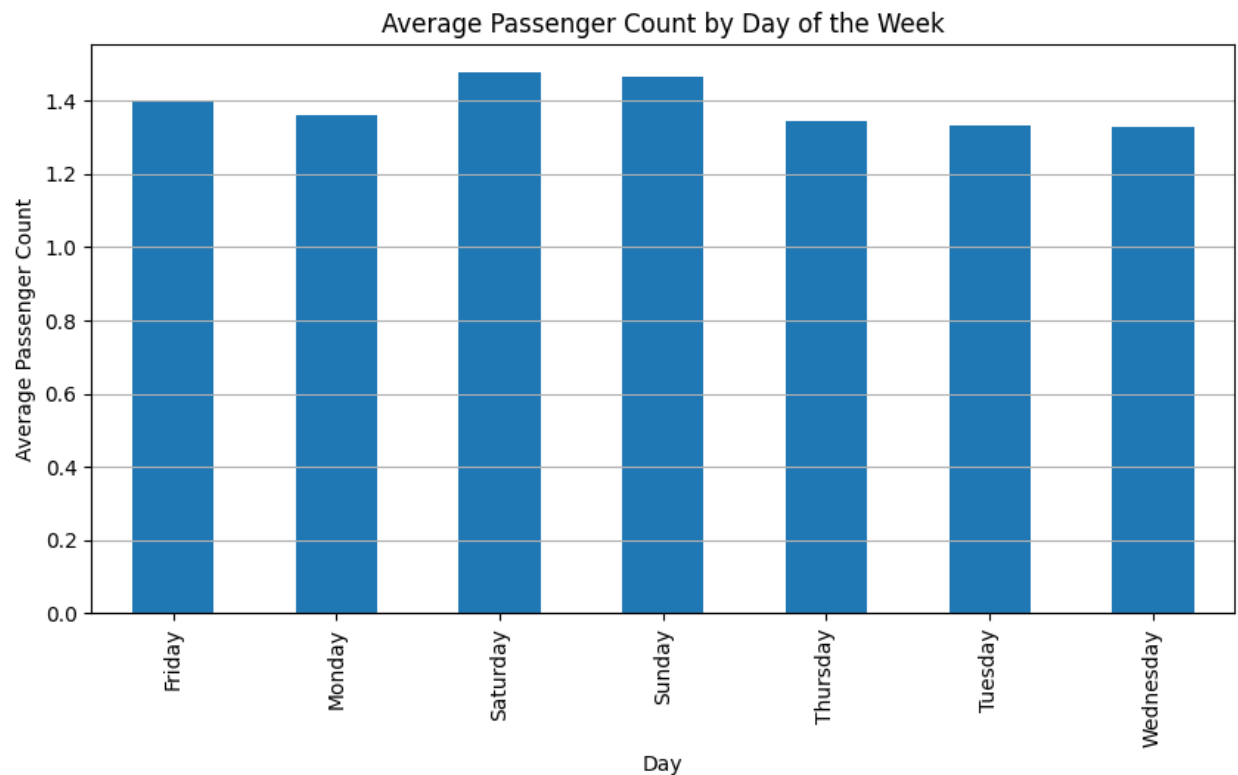
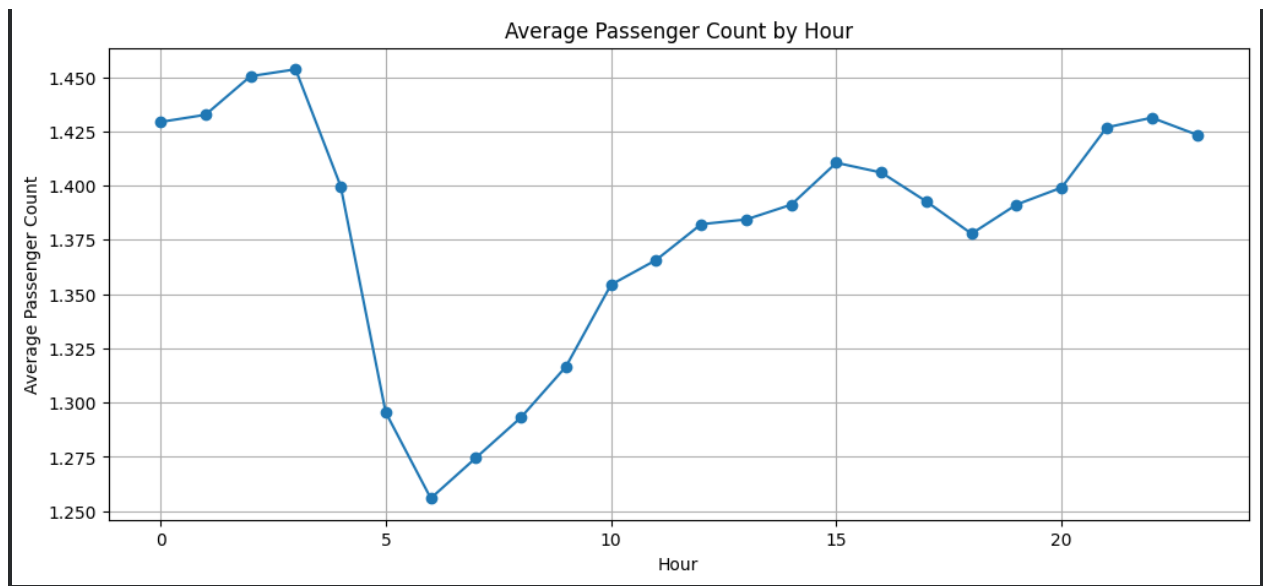
Analysis: The average fare per mile decreases as the trip distance increases for both vendors. For short trips (0–2 miles), Vendor 2 charges significantly higher fares per mile (approximately 18 USD) compared to Vendor 1 (around 10.6 USD). For medium-distance trips (2–5 miles), both vendors charge almost the same fare per mile, with Vendor 2 being slightly higher. For long trips (more than 5 miles), the fare per mile further decreases for both vendors and becomes nearly identical (around 4.5 USD per mile). This indicates that shorter trips have the highest fare per mile, while longer trips are relatively more cost-efficient. Overall, Vendor 2 charges more than Vendor 1 for short trips, whereas the difference becomes negligible as trip distance increases.

### **3.2.13. Analyse the tip percentages**



The average tip percentage varies across trip distance, passenger count, and pickup hour. Based on trip distance, shorter trips (0–2 miles) have the highest average tip percentage (22.42%), followed by medium trips (2–5 miles) at 18.77%, while longer trips (>5 miles) have the lowest average tip percentage (17.59%). This indicates that passengers tend to tip a higher percentage on shorter journeys. Across passenger counts, the average tip percentage remains relatively consistent, mostly between 19% and 21%. Four-passenger trips record the lowest average tip (17.45%), whereas six-passenger trips have the highest average tip (20.61%), suggesting that passenger count has only a minor influence on tipping behaviour. By pickup hour, tip percentages fluctuate throughout the day, remaining around 19–21% for most hours. The highest average tip percentage occurs around 5 PM (17:00) at approximately 24.38%, after which it gradually declines during the evening. Overall, trip distance has the strongest effect on tip percentage, while passenger count and pickup time show comparatively smaller variations.

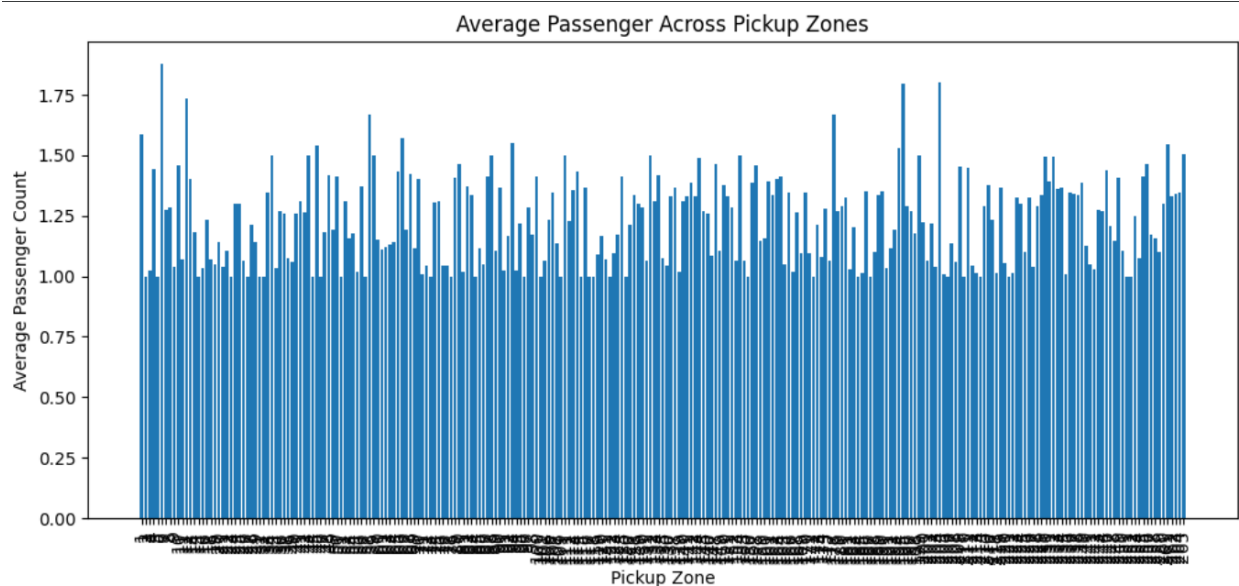
#### 3.2.14. Analyse the trends in passenger count



The average passenger count shows moderate variation across different pickup hours. Passenger counts are relatively high during the early morning hours (around 1.43–1.45 passengers per trip), decrease sharply around 5–6 AM to approximately 1.25 passengers, and then gradually increase throughout the day. Another rise is observed during the late evening (around 9–11 PM), where the average passenger count reaches about 1.42 passengers per trip. This pattern suggests higher shared travel during late-night and evening periods, while early

morning trips are more likely to involve fewer passengers. Across the days of the week, the average passenger count remains fairly consistent, ranging between approximately 1.28 and 1.45 passengers. Wednesday and Thursday record the highest average passenger counts, whereas Saturday and Sunday are slightly lower. Overall, passenger counts remain stable throughout the week, with only minor variations by hour and day.

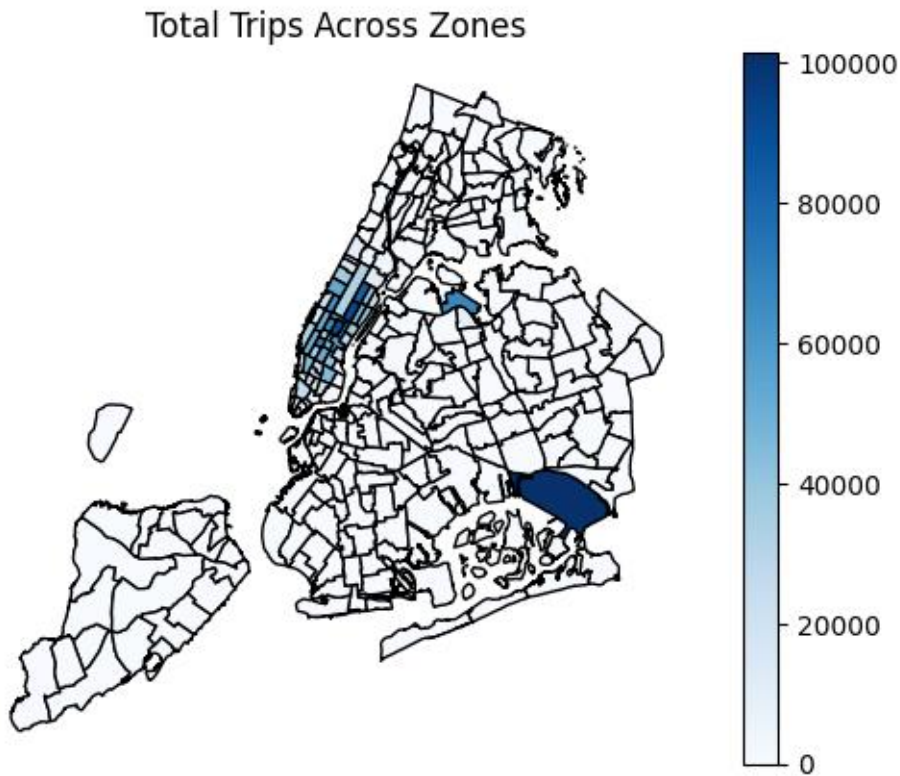
### 3.2.15. Analyse the variation of passenger counts across zones



Average Passenger Count Across Pickup Zones (Bar Chart)

#### Description:

The bar chart shows that the average passenger count varies slightly across pickup zones. Most zones have an average passenger count close to one passenger per trip, while a few zones reach approximately 1.8–1.9 passengers. This indicates that single-passenger trips are the most common across New York City, whereas a limited number of pickup zones experience relatively higher levels of shared or group travel. Overall, the variation in average passenger count across zones is moderate, with no extreme differences observed.

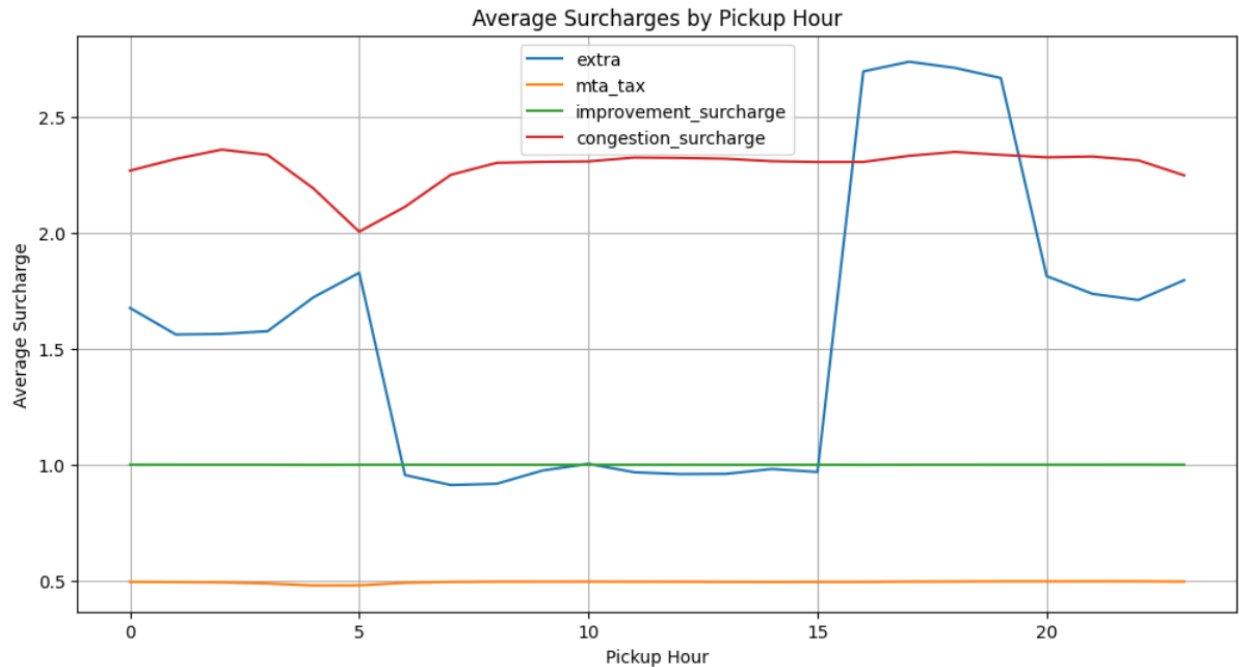


Total Trips Across Zones (Map)

Description:

The map illustrates the spatial distribution of total taxi trips across New York City taxi zones. Darker blue areas represent zones with a higher number of trips, while lighter areas indicate fewer trips. The results show that the highest trip volumes are concentrated in major commercial and transportation hubs, whereas most residential and outer zones record comparatively lower trip counts. This indicates that taxi demand is concentrated in a few high-traffic locations across the city.

**3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.**



Analysis:

The line graph shows that taxi surcharges vary throughout the day. The extra surcharge increases significantly during the evening peak hours (approximately 4 PM–8 PM), reflecting higher travel demand. The congestion surcharge remains relatively high during busy traffic periods, while the MTA tax and improvement surcharge remain almost constant across all pickup hours. Overall, the results indicate that additional charges are primarily influenced by traffic congestion and peak-hour demand rather than being applied uniformly throughout the day.

## 4. Conclusions

### 4.1. Final Insights and Recommendations

#### 4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

- Increase taxi availability during morning and evening peak hours.
- Deploy additional taxis on slow routes to reduce passenger waiting time.
- Use historical traffic patterns to optimize driver dispatch.
- Monitor high-demand pickup and dropoff zones in real time.
- Reduce idle driving by assigning nearby drivers to requests.

#### 4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

- Position more taxis in the busiest pickup zones.

Keep additional taxis near airports, railway stations, and commercial hubs.  
Reallocate taxis based on hourly demand patterns.  
Position extra taxis in high-demand night zones.  
Monitor zone-wise demand regularly to improve fleet utilisation.

**4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.**

Apply dynamic pricing during peak demand periods.  
Offer discounts during off-peak hours to increase trip demand.  
Use congestion surcharge only during heavy traffic periods.  
Regularly analyse fare and demand trends to maintain competitive pricing.  
Continuously monitor customer response and adjust pricing accordingly.